

Ratio & Proportion: Direct & Inverse



What is a Ratio?

Comparing two quantities in a simple way

A ratio compares two quantities, showing how much of one thing there is compared to another.

Example: 3 apples to 2 oranges = 3:2



3:2



Writing Ratios

Three ways to express the same comparison

- Using a colon: 3:2 (3 apples to 2 oranges)
- Using the word "to": 3 to 2
- As a fraction: $\frac{3}{2}$ or $\frac{3}{2}$

All three ways mean the same thing!



Understanding Proportion

Two ratios are proportional if they are equal

A proportion shows that two ratios are equal to each other.

Example: 2:4 is proportional to 1:2

Because $2 \div 2 = 1$ and $4 \div 2 = 2$

We write this as $2:4 = 1:2$



Direct Proportion – What is it?

When one quantity increases, the other increases at the same rate

Definition: In direct proportion, when one quantity increases, the other also increases by the same factor. Similarly, when one decreases, the other decreases too.

Example: More hours worked means more money earned!



Direct Proportion

Example

Let's solve a problem step by step!

Problem: If 4 pencils cost ₹20, how much do 6 pencils cost?

Step 1: Set up the proportion $\rightarrow 4/20 = 6/x$

Step 2: Cross multiply $\rightarrow 4 \times x = 20 \times 6$

Step 3: Solve $\rightarrow 4x = 120$, so $x = 30$

Answer: 6 pencils cost ₹30



Inverse Proportion – What is it?

When one goes up, the other goes down!

When one quantity increases, the other decreases so that their product stays constant.

Example: If more people share the same work, less time is needed to finish it!



Inverse Proportion Example

Step-by-step problem solving

Problem: 4 workers can finish a job in 12 hours. How long will 6 workers take?

Step 1: Workers $\times$ Hours = Constant

$4 \times 12 = 48$ worker-hours

Step 2: $6 \times ? = 48$

Hours = $48 \div 6 = 8$ hours

Answer: 6 workers will take 8 hours!



Direct vs Inverse Proportion

Understanding the Key Differences

Direct Proportion

- Both quantities change in the same direction
- When one increases, the other increases
- When one decreases, the other decreases
- Example: More pencils = More cost
- Formula: $x_1/y_1 = x_2/y_2$

Inverse Proportion

- Quantities change in opposite directions
- When one increases, the other decreases
- Their product always stays constant
- Example: More workers = Less time
- Formula: $x_1 \times y_1 = x_2 \times y_2$



Real-world Applications

Ratios & Proportions in Daily Life

Cooking Recipes

When you follow a recipe, you use ratios! If a recipe needs 2 cups of flour for 1 cup of sugar, the ratio is 2:1. Double the recipe? Use 4:2!

Sharing Money Fairly

If three friends earn ₹300 together and want to share in ratio 1:1:1, each gets ₹100. Proportion helps divide fairly!

Speed & Time Calculations

Inverse proportion in action! If you cycle faster, you reach school in less time. Speed increases, time decreases.

Map Reading

Maps use scale ratios like 1:1000. This means 1 cm on the map equals 1000 cm in real life!



Practice Time!

Let's practice what we learned!

Now it's your turn to solve some ratio and proportion problems.
Use the concepts of direct and inverse proportion we just
learned. Take your time and show your working!



Practice Problem 1

Direct Proportion Practice

If 5 notebooks cost 150 rupees, how much will 8 notebooks cost?
Hint: Use the direct proportion method to find your answer!



Practice Problem 2

Inverse Proportion Challenge

3 machines take 10 hours to finish a job. How long will 5 machines take?

Hint: Remember, more machines means less time needed!



Practice Problem 3

Ratio and proportion combined problem

The ratio of water to juice in a drink is 2:3. If you have 10 liters of juice, how much water is needed?



Problem 1: 5 notebooks = ₹150, so 1 notebook = ₹150 ÷ 5 = ₹30.
Therefore, 8 notebooks = ₹30 × 8 = ₹240



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Problem 2: 3 machines × 10 hours = 30 machine-hours. So 5 machines take $30 \div 5 = 6$ hours.
Problem 3: Water:Juice = 2:3. If juice = 10L, then
water = $(2/3) \times 10 = 6.67$ liters

Practice Problems Answers

Step-by-step solutions to check your work!



An illustration of three children of different ethnicities (two boys and one girl) flying kites in a grassy field with yellow flowers. The boy on the left is wearing a yellow shirt and blue shorts, holding a red kite. The girl in the middle is wearing a red shirt and orange shorts, holding an orange kite. The boy on the right is wearing an orange shirt and white shorts, holding a blue kite. They are all smiling and looking up at their kites. The background shows rolling green hills under a light blue sky.

Thank You! Any Questions?

Keep practicing and have
fun with maths!